

An LLM-Based Approach for Clarifying the Decisions of Vision Models in Autonomous Vehicles

Omid Mosalmani

Department of Computer Engineering
Shahid Chamran University of Ahvaz
Ahvaz, Iran
omid.mosalmanii@gmail.com

Mohammad Javad Rashti

Department of Computer Engineering
Shahid Chamran University of Ahvaz
Ahvaz, Iran
mohammad.rashti@scu.ac.ir

Seyed Enayat Alavi

Department of Computer Engineering
Shahid Chamran University of Ahvaz
Ahvaz, Iran
se.alavi@scu.ac.ir

Abstract—With the increasing utilization of autonomous vehicles, the transparency and explainability of their decisions have become crucial for gaining user trust and enhancing road safety. Current textual explanation methods rely on limited datasets, leading to repetitive and superficial explanations. This research presents a hybrid system where the ADAPT decision-making model is used to predict driving actions, and its attention maps serve as an interface between visual data and the explanation module. Subsequently, large language models, from the Gemini and GPT families, receive the final decision, the attention map, and a carefully designed prompt to generate concise and understandable textual explanations. The primary innovation of this approach lies in combining the decision-making model with LLMs, leveraging their extensive knowledge beyond the constraints of training data to enable the generation of more precise and diverse explanations. The system is evaluated on the BDD-X dataset and measured against standard captioning metrics including BLEU-4, METEOR, ROUGE-L, CIDEr-D, and SPICE. The evaluation results indicate the superiority of explanation outputs in our system, compared to the baseline ADAPT, particularly in multi-reference scenarios, providing more fluent and contextually rich explanations. For instance, the output acquired from Gemini 2.5 Pro model achieves a METEOR score of approximately 19.45, a significant improvement of about 28 percent compared to 15.2 for ADAPT. Furthermore, supplementary experiments show that using a contour representation of the attention map and fine-tuning the models lead to increased visual-textual consistency and result stability. In summary, by linking the visual attention of the decision-making model to the linguistic capabilities of LLMs, this research takes a step toward developing more explainable and trustworthy autonomous vehicles.

Keywords—Explainable AI, Prompt Engineering, Large Language Models, Autonomous Vehicles, Textual Explanation

I. INTRODUCTION

Recent advances in deep learning have enabled autonomous vehicles to perceive complex environments and make real-time decisions. Despite their success, these systems often operate as “black boxes,” where neither developers nor users have clear access to the underlying reasoning behind model outputs. This lack of transparency becomes critical in autonomous driving, where system decisions need to be evaluated based on the regional AI acts, or directly impact human safety.

To address this challenge, the field of Explainable Artificial Intelligence (XAI) has introduced methods such as

Grad-CAM[1], LIME[2], and SHAP[3], which provide visual or numerical explanations of model behavior. However, the results of these approaches are primarily designed for technical experts and are not easily interpretable by everyday users. Thus, generating human-understandable textual explanations has emerged as an important step toward improving human-machine interaction.

Large Language Models (LLMs) such as ChatGPT and Google Gemini have recently been explored as tools for producing natural language explanations. By leveraging prompt engineering, these models can translate complex decision-making processes into fluent human language [4]. In the context of autonomous driving, they offer a way to improve user trust by clarifying system behavior. Nevertheless, LLMs alone face limitations: they generally lack direct access to perception models’ internal reasoning and may instead generate plausible—but not necessarily accurate—descriptions based on linguistic patterns.

This study introduces a hybrid framework that combines an action-aware driving model (ADAPT[5]) with LLMs to enhance interpretability. The ADAPT model produces both driving actions and attention maps, which highlight visual regions influencing the decision. These outputs are then integrated into LLMs through carefully designed prompts, enabling the generation of short, natural, and semantically aligned explanations. The novelty of this approach lies in combining visual attention mechanisms with LLMs’ linguistic capabilities, producing explanations that are both fluent and faithful to the actual decision-making process.

The proposed system is evaluated on the BDD-X dataset [6] using established captioning metrics such as BLEU-4, METEOR, ROUGE-L, CIDEr-D, and SPICE. Results demonstrate that the hybrid method outperforms the ADAPT baseline, particularly under single-reference evaluation. For example, Gemini 2.5 Pro achieved a METEOR score of approximately 19.45 compared to 15.2 for ADAPT, representing a 28% improvement. Additional experiments using contour-based visualization of attention maps and fine-tuning strategies show further increased alignment between visual cues and textual explanations.

In summary, this work contributes a novel framework that links the visual attention of driving decision models with the expressive power of LLMs. The findings suggest that the proposed approach can enhance both transparency and trust in autonomous vehicles, paving the way for future research on multimodal explainability in safety-critical systems.

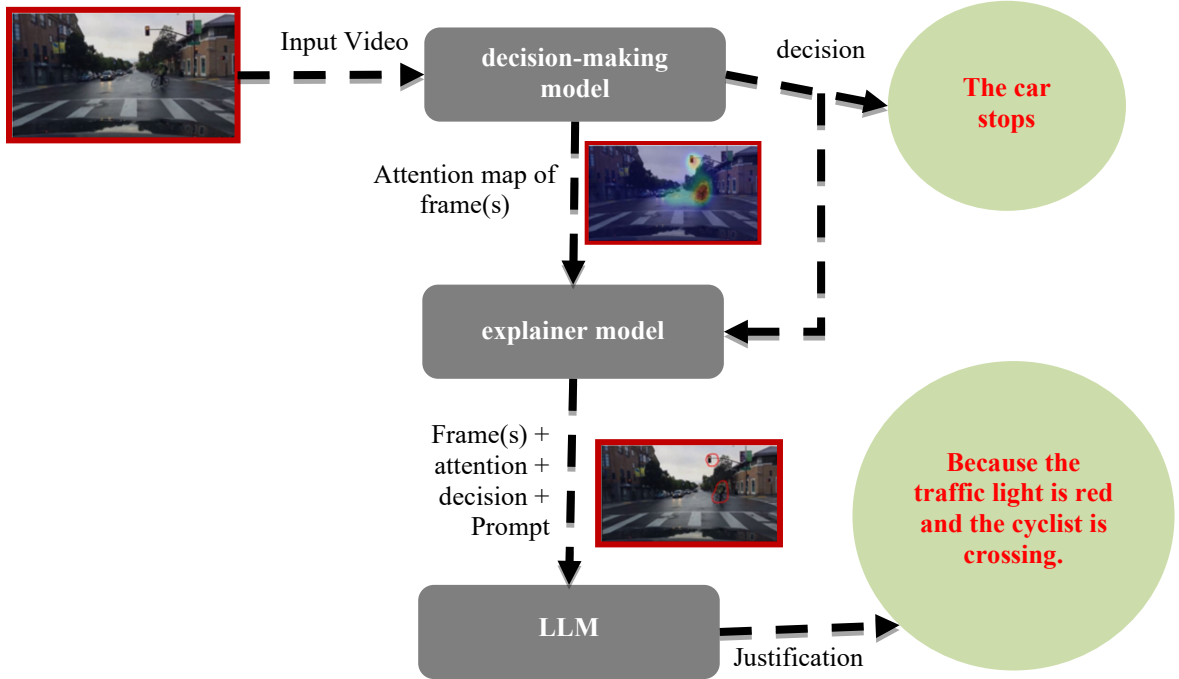


Fig. 1: Block diagram of the proposed system architecture (red parts are examples)

II. RELATED WORK

Early research in explainable artificial intelligence (XAI) for computer vision has focused primarily on visual explanation methods such as Grad-CAM, LIME, and SHAP. These techniques aim to highlight image regions that most influence a model’s decision. For instance, Grad-CAM generates class-specific saliency maps using gradient information, LIME approximates model behavior with a locally interpretable linear model, and SHAP assigns feature importance scores based on Shapley values [2, 3]. While these approaches provide valuable insights into the internal structure of deep neural networks, their outputs—such as heatmaps or feature importance plots—are often difficult for non-experts to interpret, limiting their practical utility.

More recently, the use of LLMs for explainability in autonomous driving has gained attention. Chen et al. [7] proposed a model that integrates object-level features from driving scenes with an LLM to generate natural language explanations of road situations and corresponding decisions. Similarly, systems such as DriveGPT4 [8] have been designed to jointly predict driving actions and provide textual justifications by processing multi-frame visual inputs along with user queries. These developments demonstrate that LLMs can enhance action transparency in autonomous vehicles by producing explanations that are more natural and comprehensible for human users. Survey studies further confirm the importance of XAI in this domain, highlighting a trend toward text-based explanations supported by the generative capabilities of LLMs [9].

Beyond autonomous driving, other works have applied LLMs to explainability in computer vision. Zhao et al. [10] introduced a post-processing framework that automatically generates textual explanations for individual neurons in deep visual networks. Another approach, LVX[11], leverages an LLM as an “explainer” to produce hierarchical visual attributes, thereby providing semantic interpretations of internal model concepts. These studies indicate that LLMs,

by drawing on their broad conceptual knowledge, can produce richer and more accurate interpretations of vision models.

Despite these advances, several limitations remain. Many visual explanation techniques, such as Grad-CAM and LIME, are tailored for expert audiences and remain inaccessible to lay users [1, 2, 12]. Meanwhile, current LLM-based approaches often neglect the internal mechanisms of vision models, particularly attention maps, which contain crucial cues about the decision-making process. To bridge this gap, the method proposed in this paper combines attention maps generated by an action-aware model (ADAPT [5]) with LLM-based textual explanation. This hybrid approach aims to provide both explicit visual grounding and fluent, human-understandable reasoning, thereby enhancing the interpretability of autonomous driving systems.

Attention maps have been investigated as intermediaries between visual recognition and language-based explanation models. For example, DriveBLIP2 (Ling et al. 2025) augments a BLIP2-OPT vision-language model with an auxiliary attention-map generator that highlights salient driving-scene objects and feeds the focused features into the LLM’s decoder to produce more targeted textual descriptions [13]. Nevertheless, generating highly accurate, decision-specific explanations remains an open challenge; in contrast to DriveBLIP2’s object-centric strategy, the present work employs an LLM to directly clarify the vision model’s overall decision rationale.

Recent work has explored using multimodal LLMs to generate human-readable rationales for driving decisions. For example, DriveGPT4 fine-tunes a multimodal language model on a curated driving dataset, enabling it to jointly predict vehicle control signals and produce natural-language action descriptions and justifications [8]. However, the authors note that when trained only on driving-specific instruction data, DriveGPT4 may hallucinate—for example, detecting non-existent vehicles or traffic lights—which undermines reliability in open or novel driving scenes [8]. To solve this problem, the method RAG-Driver augments a

driving LLM with retrieval-augmented in-context learning: it retrieves analogous expert-driving demonstrations to ground its outputs, yielding state-of-the-art driving action explanations and strong zero-shot generalization to unseen scenarios. RAG-Driver improves generalization via retrieval, but it still exhibits occasional hallucinations and relies on a relatively small LLM backbone (7B parameters) for feasibility [14]. Also, prior studies indicate that fine-tuning tends to yield stronger performance in broad, open-domain tasks—such as those encountered in autonomous driving—where extensive adaptation of the model’s internal representations is beneficial, while retrieval-augmented generation (RAG) is particularly effective in tasks requiring access to specialized or external knowledge beyond an LLM’s pre-training corpus [15-17].

III. PROPOSED METHOD

A. Overall Architecture

The proposed system consists of three main components (Fig. 1). First, a vision encoder based on the ADAPT architecture [5] processes input video frames and predicts the corresponding driving action. ADAPT, an action-aware transformer, not only generates the driving decision but also produces spatial attention maps that indicate which image regions influenced the output.

In the second stage, an attention map extraction module processes these outputs. By applying thresholding and edge detection, the system converts the raw attention maps into contour-based visualizations that highlight the most influential regions.

Finally, an LLM-based explanation module receives the contour map, the driving decision predicted by ADAPT (e.g., “brake” or “turn left”), and a carefully designed prompt. Using this multimodal input, the LLM (e.g., GPT-4o or Gemini) produces short, natural-language explanations of the decision. The design ensures that visual attention cues and linguistic knowledge are combined in a coherent manner, resulting in explanations that are both interpretable and faithful to the decision-making process.

B. Attention Map Extraction and Representation

The attention maps generated by ADAPT are continuous-valued matrices that indicate the proportion of model focus at each spatial location. To make this information interpretable by the LLM, we apply a transformation pipeline. First, a threshold (e.g., the top 10% of attention values) is applied to isolate key regions. Next, edge detection is performed to extract contour representations of these regions. The result is a binary contour representation, visualized as red outlines that explicitly mark the most influential visual areas. This contour image is then used as part of the multimodal input to the LLM. By providing both textual and visual cues, the model gains a more explicit grounding in the evidence that influenced the driving decision.

C. Prompt Engineering

Fig. 2 presents the prompt used in this research. The prompt used across all experiments is fixed and carefully designed to guide the LLM in producing concise explanations of driving scenarios. Each prompt includes

You are a description system in driving scenarios. Your task is to capture an image of a driving scenario, the decision, and the area of interest of the decision-making system. Then explain why the decision-making system made that decision. Your input is {{frames_count}} frames of a camera in front of the vehicle, including the area of interest marked with a red circle. Explain simply, briefly, and without explanation in only 1 or 2 short sentences. Avoid additional sentences, requests for more information, etc., do not explain context or given input information. You do not have access to the full information that the decision-making system does. So you consider that decision correct. Under no circumstances should you write more than 2 sentences. Avoid writing sentences like 'Based on the provided image ...' or 'In conclusion ...'. Here are some examples:
 * since the truck in front wasn't moving fast.
 * because the light is green.
 * due to a vehicle in front of it moving slowly as it turns right.

Fig. 2: Full text of the prompt used to generate the explanation (output of the explanation model)

three elements: (1) a persona definition that assigns the role of a “description system” in driving contexts to the model, (2) contextual inputs consisting of video frames, the driving action, and the corresponding attention region, and (3) explicit output constraints and short few-shot examples.

The constraints require the model to produce explanations in one or two short sentences, avoiding additional commentary, context restatements, or irrelevant details. The system assumes that the decision made by ADAPT is correct and explains it accordingly. Example outputs include simple, natural justifications such as “since the truck in front wasn’t moving fast” or “because the light is green.”

This prompt design serves three practical goals: (1) reducing hallucinations by narrowing the scope of responses, (2) ensuring consistency across samples by maintaining a fixed prompt, and (3) producing comparable, concise explanations suitable for both human evaluation and automatic metrics. All generation parameters (e.g., temperature, max tokens) are recorded to guarantee reproducibility of experiments.

D. Fine-Tuning

Fine-tuning refers to the process of adapting a pre-trained language model using a domain-specific dataset, allowing its parameters to be adjusted to better capture the style and requirements of the target task. This typically improves the accuracy and relevance of generated outputs but requires significant computational resources and large-scale annotated data.

In this study, fine-tuning is performed on two cloud platforms: Gemini 2.5 Pro on Google Cloud and GPT-4o on the OpenAI platform. The training dataset was derived from the BDD-X dataset, ensuring consistency across experiments. Hyperparameter configurations were handled

automatically by each platform, with pipeline management—including data upload, training execution, and artifact storage—fully managed by the cloud services.

The primary objective of fine-tuning is to align the model outputs with the constraints of the designed prompt (i.e., short, one–two sentence explanations). Final results are evaluated using the same captioning metrics as earlier stages, including BLEU, METEOR, CIDEr, and SPICE, to allow a direct comparison between baseline and fine-tuned configurations. Recording platform parameters and reusing the same dataset ensures a high degree of reproducibility.

E. Explanation Generation Pipeline

The entire explanation generation process can be summarized in several sequential steps:

- **Input:** Video frames from the vehicle’s front camera are processed by the ADAPT model to produce both a predicted driving action and a raw attention map.
- **Pre-processing:** Input video frames are temporally downsampled and spatially center-cropped to meet the model’s input requirements. The attention map is then refined through thresholding and contour extraction.
- **Prompt Construction:** A prompt is generated that integrates the predicted action and contour-based attention visualization along with the task-specific instructions.
- **LLM Processing:** The constructed prompt is passed to the LLM (e.g., GPT-4o or Gemini), which processes both the visual and textual cues to produce a natural-language explanation of the decision.

The outcome of this pipeline is a concise and human-readable explanation that reflects both the visual context of the driving scene and the decision predicted by ADAPT, enriched with the linguistic knowledge of LLMs.

IV. DATA AND EVALUATION METRICS

A. Dataset

The experiments in this study were conducted using the Berkeley DeepDrive Explanation (BDD-X) dataset[6], which is part of the larger Berkeley DeepDrive project [18]. BDD-X contains over 77 hours of driving videos, for a total of 6,970 clips. Each driving scenario in the dataset is annotated with a natural language explanation consisting of both the action (e.g., “the car stops”) and the reason (e.g., “because the traffic light is red”).

TABLE I. BDD-X DATASET STATISTICS

<i>Split</i>	<i>Number of Clips</i>
Training	21,143
Validation	2,519
Test	2,566
Total	26,228

For preprocessing, each video is segmented into fixed-length samples of 32 frames, which serves as the model’s input unit. The final dataset comprises of 26,228 clips, divided into 21,143 training samples, 2,519 validation

samples, and 2,566 test samples. TABLE I summarizes the dataset statistics. The use of the official BDD-X release ensures high-quality annotations and standardized evaluation settings.

B. Automatic Evaluation Metrics

To quantitatively evaluate the generated explanations, we adopt some of the established captioning metrics widely used in natural language generation. These metrics capture both lexical overlap and semantic alignment:

- **BLEU-4:** Measures the precision of 4-gram overlaps between generated and reference texts, emphasizing local word accuracy [19].
- **ROUGE-L:** Focuses on the longest common subsequence, reflecting fluency and coverage of essential content [20].
- **METEOR:** Incorporates synonym matching and stemming, offering a more flexible measure of linguistic quality and semantic similarity [21].
- **SPICE:** Evaluates semantic content by parsing objects, attributes, and relationships in the explanation and comparing them with reference annotations [22].
- **CIDEr-D:** Weighs n-gram overlaps using TF-IDF, rewarding explanations that align with human references while penalizing generic outputs [23].

By combining word-level and semantic-level evaluation, this metric set provides a comprehensive assessment of explanation quality, covering accuracy, fluency, and conceptual relevance.

V. EXPERIMENTAL SETUP

The experiments were conducted on the Google Colab environment. Most vision model executions used NVIDIA A100 GPUs with 40 GB memory, while L4 GPUs were occasionally employed. The software stack included CUDA 12.4 and PyTorch 2.6. Code version control was maintained through GitHub, with commit IDs and execution configurations recorded to ensure reproducibility.

Four multimodal variants of language models are evaluated: Gemini 2.5 Pro, Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, and GPT-4o (a compact variant of GPT-5). In the final two stages, Gemini 2.5 Pro and GPT-4o are fine-tuned on the BDD-X dataset for domain-specific adaptation.

Baseline training parameters for ADAPT, the core decision-making model, are summarized in TABLE II. Key settings include a training batch size of 4, learning rate of 0.0002, 40 epochs, a backbone learning rate of 0.0005, masking probability of 0.5, and gradient clipping of 1. Training was conducted using AdamW optimization with mixed-precision (fp16) under DeepSpeed. No early stopping was applied.

TABLE II. KEY ADAPT TRAINING PARAMETERS

<i>Parameter</i>	<i>Value</i>
Training batch size	4
Validation batch size	12
Frames per clip	32

<i>Parameter</i>	<i>Value</i>
Learning rate	0.0002
Backbone learning rate	0.0005
Epochs	40
Masking probability	0.5
Gradient clipping	1

Three baseline configurations are considered for comparison:

- 1) **ADAPT-only**: Decisions generated without LLMs.
- 2) **ADAPT + LLM (Prompt-only)**: Integration of LLMs using few-shot prompt examples.
- 3) **ADAPT + LLM (Fine-tuned)**: LLMs fine-tuned on the BDD-X dataset for domain alignment.

These baselines served as reference points against the proposed hybrid approach.

VI. QUANTITATIVE RESULTS

A. Main Comparison

The primary results across the five metrics (BLEU-4, METEOR, ROUGE-L, CIDEr-D, SPICE) are presented in TABLE III. Several key trends were observed:

- **Semantic improvements with LLMs**: Compared to ADAPT-only outputs, configurations involving LLMs showed significant gains in semantic-oriented metrics such as METEOR and SPICE. This indicates that LLMs are better suited for expressing causal relations and object references in human-readable form.
- **Lexical overlap sensitivity**: BLEU-4 and CIDEr-D scores were generally higher for GPT-family models, particularly when fine-tuned, reflecting closer word-level similarity to reference texts. However, higher lexical overlap does not necessarily imply superior semantic faithfulness.
- **Illustrative example**: Gemini 2.5 Pro in its base form achieved BLEU-4 = 7.01 and SPICE = 24.35, while fine-tuned configurations demonstrated marked improvements in ROUGE-L and METEOR scores, highlighting the benefits of domain adaptation.

TABLE III. BLEU-4, METEOR, ROUGE-L, CIDEr AND SPICE BENCHMARK VALUES FOR THE BASIC (ADAPT AND DRIVEGPT4) AND COMBINED (ADAPT+LLM) METHODS

<i>model</i>	<i>BLEU4</i>	<i>CIDEr</i>	<i>METEOR</i>	<i>ROUGE-L</i>	<i>SPICE</i>
ADAPT	11.04	102.6	15.2	32	25.6
DriveGPT 4	10.02	98.71		31.52	
Gemini 2.5 Pro	7.01	58.47	17.72	32.77	24.35
Gemini 2.5 Flash	5.64	51.28	16.84	31.75	20.06
Gemini 2.5 Flash Lite	6.59	52.56	15.33	29.81	20.51
GPT 5 Mini	2.03	25.48	14.01	23.29	11.17

<i>model</i>	<i>BLEU4</i>	<i>CIDEr</i>	<i>METEOR</i>	<i>ROUGE-L</i>	<i>SPICE</i>
FT-GPT 4o	11.09	103	15.24	33.84	26.25
FT-Gemini 2.5 Pro	11.83	114.73	19.45	40.57	28.47

B. Prompt-only vs. Fine-Tuning

A direct comparison between prompt-only and fine-tuned regimes reveals clear distinctions:

- **Prompt-only models**: produce linguistically coherent explanations but often fail to capture fine-grained reasoning, such as specific causal triggers or object prioritization. This is reflected in moderate METEOR and SPICE scores.
- **Fine-tuned models** demonstrate stronger alignment with domain-specific terminology and visual cues, resulting in significant gains in ROUGE-L and METEOR. Although fine-tuning requires additional computational resources and labeled data, the performance benefits justify its use in high-stakes applications.

C. Sensitivity to Context Length

Analysis of input frame lengths shows that increasing temporal context improves semantic quality. Configurations with 16 or 32 frames consistently outperformed those with 1 or 4 frames in METEOR, ROUGE-L, and SPICE. This trend is expected, as longer temporal sequences capture richer contextual dynamics and inter-object relations. Detailed comparisons are presented in TABLE IV.

D. Gemini vs. GPT Models

A cross-family comparison among LLMs indicates complementary strengths:

- **Gemini models** generally excel in semantic and visual fidelity, achieving higher SPICE and METEOR scores.
- **GPT models** (particularly fine-tuned versions) perform better on lexical overlap metrics such as BLEU and CIDEr, reflecting stronger alignment with reference phrasings.

This distinction suggests that the choice of model should be guided by application requirements—whether semantic depth or lexical precision is prioritized.

TABLE IV. EFFECT OF FRAME NUMBER (N) ON METRICS IN ADAPT BASELINE (BDD-X HUMAN JUSTIFICATION ONLY)

<i>Input (n) frames</i>	<i>BLEU4</i>	<i>CIDEr</i>	<i>METEOR</i>	<i>ROUGE-L</i>	<i>SPICE</i>
1 frame	2.79	36.23	9.99	23.44	12.78
4 frames	7.75	82.34	12.89	29.76	21.86
8 frames	8.19	81.49	13.47	31.33	23.70
12 frames	4.57	46.97	11.55	24.82	14.63
16 frames	8.70	91.50	13.80	32.0	24.80
32 frames	11.40	102.6	15.20	32.0	25.60

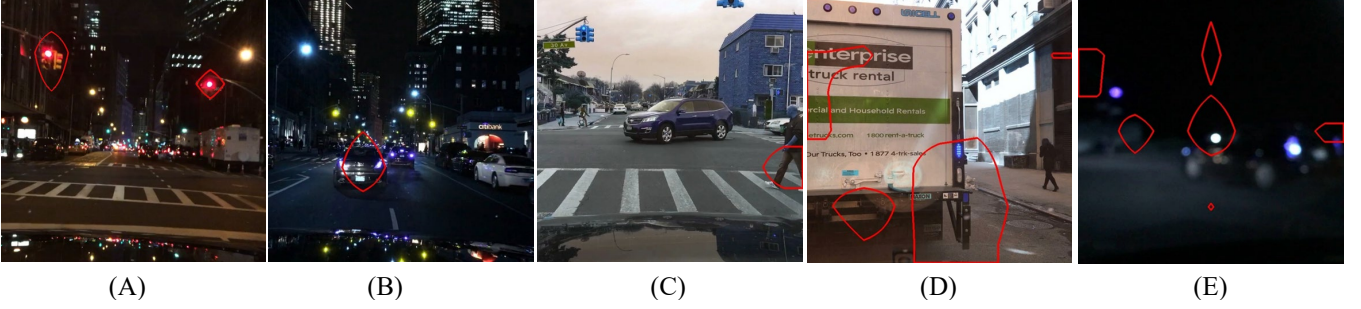


Fig. 3: Qualitative Case Studies of Explanations.

VII. ERROR ANALYSIS, ROBUSTNESS QUALITATIVE ANALYSIS AND CASE STUDIES

To complement the quantitative metrics, a qualitative assessment was conducted to evaluate the visual-textual consistency of the system. This analysis focuses on the output of fine-tuned models (FT-Gemini Pro and FT-GPT-4o) compared to the ADAPT baseline and human references. *Fig. 3* illustrates the attention contours (red mask) generated by the ADAPT model and the corresponding textual explanations produced by the fine-tuned LLMs. Cases A, B, and C demonstrate successful visual-textual grounding: A (Red Light), B (Car Following), and C (Pedestrian) show the contour accurately bounding the causal object, leading to correct and concise justifications. Cases D and E show failure modes: D (Mis-grounding), where the contour highlights a parked truck but the model hallucinates a "pedestrian crossing." E (Ambiguity), where low visibility results in a diffuse contour, leading to a generic output ("heavy traffic") instead of identifying the stopped car.

The selection of cases is based on ROUGE-L and METEOR scores, categorizing samples into high-performing (successful) and low-performing (counter-example) groups.

A. Evaluation Criteria

For each case study, the visual attention contour and the generated explanation were analyzed against a three-point checklist:

- 1) **Visual Grounding:** Is the causal object (e.g., traffic light, pedestrian) explicitly visible within the red attention contour?
- 2) **Semantic Equivalence:** Is the generated explanation semantically equivalent to the human reference, even if the wording differs?
- 3) **Conciseness:** Is the text direct and free from hallucinations or irrelevant context?

In successful cases, the system satisfied all three criteria, demonstrating strong alignment between the visual area of interest and the textual reasoning.

B. Successful Scenarios

The following examples illustrate instances where the model successfully grounded its explanation in the visual data.

Case A- Red Light / Stop: In this scenario (*Fig. 3-A*), the attention contour is focused on the traffic light and the intersection area. The driving action is deceleration or stopping.

- **FT-Gemini Pro Output:** "Since the light is red."
- **Human Reference:** "Due to red light at the intersection."
- **Analysis:** The model produces a concise explanation that is semantically identical to the reference. The text aligns perfectly with the object highlighted in the contour, showing precise visual-textual consistency.

Case B- Car Following: The contour in this example (*Fig. 3-B*) highlights the leading vehicle and the longitudinal distance.

- **FT-GPT-4o Output:** "Because the vehicle in front is moving slowly."
- **Human Reference:** "Since the car in front has low speed."
- **Analysis:** The explanation directly references the causal factor (the car) contained within the contour. There are no hallucinations, and the reasoning supports the action of maintaining distance.

Case C- Pedestrian Crossing: In this case (*Fig. 3-C*), the attention map highlights a pedestrian within the crosswalk.

- **FT-Gemini Pro Output:** "For pedestrian crossing."
- **Human Reference:** "Due to presence of pedestrian in crosswalk."
- **Analysis:** The explanation is brief and accurate. The visual focus on the pedestrian is correctly translated into a justification for yielding, demonstrating safe and justifiable reasoning.

C. Failure Modes and Counter-Examples

Counter-examples reveal specific weaknesses, primarily related to visual-textual mismatch (mis-grounding) or environmental ambiguity.

Case D- Visual-Textual Mismatch: In this instance (*Fig. 3-D*), the contour highlights a parked delivery truck narrowing the path. The action is "slow down".

- **Model Output:** "For pedestrian crossing."
- **Human Reference:** "Because the car in front narrows the path."
- **Analysis:** This represents a classic "mis-grounding" error. The model hallucinates a pedestrian—a common high-priority safety class—despite the contour clearly showing a vehicle. The text refers to an object not present in the visual evidence.

Case E- Environmental Ambiguity: This night-time scene (Fig. 3-E) features low contrast, scattered light reflections, and a diffuse attention contour.

- **Model Output:** "Since traffic is heavy."
- **Human Reference:** Due to the car in front stopping."
- **Analysis:** Due to visual ambiguity and low visibility, the model resorts to a generic, safe guess ("heavy traffic") rather than identifying the specific stopped vehicle. This suggests that when visual cues are weak, the model relies on generalized priors rather than specific visual evidence.

VIII. DISCUSSION

Semantic analysis of the quantitative results suggests that models benefiting from either domain-specific fine-tuning or explicit visual constraints (contour-based attention maps) consistently achieved higher scores in semantic-oriented metrics such as METEOR and SPICE. For instance, baseline multimodal models in the prompt-only setting achieved BLEU-4 ≈ 7 and SPICE ≈ 24 , while fine-tuned variants improved to BLEU-4 ≈ 11 with substantial gains in SPICE ≈ 28 and ROUGE ≈ 41 . This demonstrates the role of fine-tuning in aligning vocabulary and style with domain-specific explanations.

Attention maps also play a critical role in enhancing explanation fidelity. When important image regions are explicitly provided as contour overlays, models produce fewer irrelevant references, resulting in higher semantic alignment scores. The attention maps effectively act as a *visual grounding constraint*, guiding the LLM toward more accurate and context-specific reasoning.

Nevertheless, practical considerations remain significant. Fine-tuning large-scale LLMs requires high-memory GPUs, long training cycles, and significant storage, which makes real-time deployment in autonomous driving impractical [24]. Parameter-efficient methods such as Low-Rank Adaptation (LoRA) reduce these costs by freezing the base model and training small low-rank adapter matrices, lowering memory usage and the number of trainable parameters by orders of magnitude [24, 25]. Because LoRA adapters can be merged back into the base model, inference latency is negligible compared to full fine-tuning [24]. However, LoRA still introduces engineering overhead—such as selecting adapter ranks and integration into pipelines—and managing multiple adapters can increase operational complexity in real-time systems [25]. Although LoRA reduces trainable parameters dramatically (e.g., 18M vs. 175B for GPT-3 [24], or $<1\%$ for 65B models [25]), deploying multiple adapters (e.g., 10+) can still add hundreds of millions of parameters in aggregate, increasing memory footprint by several gigabytes and introducing measurable latency overhead in real-time pipelines.

Ethical and risk-related concerns are also noteworthy. Two major risks are: (1) **hallucinations**, where LLMs introduce fabricated or unverifiable details, and (2) **false trust**, where users over-rely on explanations that may not be fully accurate. To address these risks, practical safeguards are recommended:

- **Post-hoc verification:** Cross-check explanations against attention maps and extracted features, flagging mismatched outputs.
- **Confidence reporting:** Accompany explanations with uncertainty labels (e.g., "low confidence").
- **Human-in-the-loop:** In high-risk safety-critical decisions, human review and logging of explanations should be incorporated.
- **Scope limitation:** Restrict explanations to verifiable content, penalizing hallucinations during training.

Overall, combining visual constraints (attention maps) with fine-tuned LLMs appears to be the most effective strategy for producing fluent, semantically faithful, and trustworthy explanations. However, practical deployment will require careful engineering to balance quality, cost, latency, and ethical safeguards.

IX. CONCLUSION AND FUTURE WORK

This paper introduces a framework for generating natural language explanations of autonomous driving decisions by integrating an attention-aware vision model (ADAPT) with large language models. Quantitative results demonstrate that LLMs, particularly when fine-tuned, significantly improve the semantic quality and fluency of explanations. Incorporating contour-based attention maps further enhances fidelity to visual evidence, reducing irrelevant references and increasing semantic alignment.

The contributions of this study are threefold:

- 1) Showing the effectiveness of combining attention-aware vision with LLMs for human-understandable explanations.
- 2) Providing a comparative analysis of prompt-only versus fine-tuned regimes, highlighting the significance of fine-tuning for improved explainability.
- 3) Presenting error analysis and proposing risk-mitigation strategies to address hallucinations and false trust.

Future directions include:

- Incorporating **temporal attention mechanisms** to capture scene dynamics more effectively.
- Conducting **structured human evaluations** (e.g., ≥ 200 samples with multiple annotators) to measure correlation between human judgment and automatic metrics.
- Developing **hallucination-reduction techniques**, such as adversarial training, response filtering, or stronger grounding mechanisms.
- Testing the system in **field scenarios** to assess real-time latency, robustness, and stability.
- Exploring **model compression approaches** (e.g., quantization, distillation, LoRA) for deployment on resource-constrained in-vehicle hardware.

By advancing explainability with multimodal reasoning, this framework paves the way for more transparent and trustworthy autonomous driving systems, while providing a foundation for future work in multimodal XAI.

ACKNOWLEDGMENT

This study was supported in part by Shahid Chamran University of Ahvaz grant # SCU.EC1403.30899.

REFERENCES

- [1] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 618–626.
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "" Why should i trust you?" Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
- [3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in neural information processing systems*, vol. 30, 2017.
- [4] X. Amatriain, "Prompt design and engineering: Introduction and advanced methods," *arXiv preprint arXiv:2401.14423*, 2024.
- [5] B. Jin *et al.*, "Adapt: Action-aware driving caption transformer," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023: IEEE, pp. 7554–7561.
- [6] J. Kim, A. Rohrbach, T. Darrell, J. Canny, and Z. Akata, "Textual explanations for self-driving vehicles," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 563–578.
- [7] L. Chen *et al.*, "Driving with llms: Fusing object-level vector modality for explainable autonomous driving," *arXiv preprint arXiv:2310.01957*, 2023.
- [8] Z. Xu *et al.*, "Drivegpt4: Interpretable end-to-end autonomous driving via large language model," *IEEE Robotics and Automation Letters*, 2024.
- [9] S. Atakishiyev, M. Salameh, H. Yao, and R. Goebel, "Explainable artificial intelligence for autonomous driving: A comprehensive overview and field guide for future research directions," *IEEE Access*, 2024.
- [10] C. Zhao, W. Qian, Y. Shi, M. Huai, and N. Liu, "Automated Natural Language Explanation of Deep Visual Neurons with Large Models," *arXiv preprint arXiv:2310.10708*, 2023.
- [11] X. Yang and X. Wang, "Language model as visual explainer," *Advances in Neural Information Processing Systems*, vol. 37, pp. 135094–135128, 2024.
- [12] P. Mavrepis, G. Makridis, G. Fatouros, V. Koukos, M. M. Separdani, and D. Kyriazis, "XAI for All: Can Large Language Models Simplify Explainable AI?," *arXiv preprint arXiv:2401.13110*, 2024.
- [13] S. Ling, Y. Wan, X. Jia, and N. Du, "DriveBLIP2: Attention-Guided Explanation Generation for Complex Driving Scenarios," *arXiv preprint arXiv:2506.22494*, 2025.
- [14] J. Yuan *et al.*, "Rag-driver: Generalisable driving explanations with retrieval-augmented in-context learning in multi-modal large language model," *arXiv preprint arXiv:2402.10828*, 2024.
- [15] S. Alghisi, M. Rizzoli, G. Roccabruna, S. M. Mousavi, and G. Riccardi, "Should we fine-tune or rag? evaluating different techniques to adapt llms for dialogue," *arXiv preprint arXiv:2406.06399*, 2024.
- [16] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, "Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs," Miami, Florida, USA, November 2024: Association for Computational Linguistics, in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 237–250, doi: 10.18653/v1/2024.emnlp-main.15. [Online]. Available: <https://aclanthology.org/2024.emnlp-main.15/>
- [17] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in neural information processing systems*, vol. 33, pp. 9459–9474, 2020.
- [18] F. Yu *et al.*, "Bdd100k: A diverse driving dataset for heterogeneous multitask learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 2636–2645.
- [19] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, 2002, pp. 311–318.
- [20] C.-Y. Lin and F. J. Och, "Automatic evaluation of machine translation quality using longest common subsequence and skip-bigram statistics," in *Proceedings of the 42nd annual meeting of the association for computational linguistics (ACL-04)*, 2004, pp. 605–612.
- [21] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, 2005, pp. 65–72.
- [22] P. Anderson, B. Fernando, M. Johnson, and S. Gould, "Spice: Semantic propositional image caption evaluation," in *European conference on computer vision*, 2016: Springer, pp. 382–398.
- [23] R. Vedantam, C. Lawrence Zitnick, and D. Parikh, "Cider: Consensus-based image description evaluation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 4566–4575.
- [24] E. J. Hu *et al.*, "Lora: Low-rank adaptation of large language models," *ICLR*, vol. 1, no. 2, p. 3, 2022.
- [25] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "Qlora: Efficient finetuning of quantized llms," *Advances in neural information processing systems*, vol. 36, pp. 10088–10115, 2023.